Sensing-Driven Meal Guidance Across the Menstrual Cycle: Forecasting Symptoms and Glucose Instability from Consumer Wearables

COMS 6998 Mobile Health: Sensing, AI, and Human-Centered Design, Project Proposal

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1. Problem

Many menstruating people experience cycle-linked fatigue, cravings, cramps, and bloating. "Cycle syncing" content and commercial platforms claim that diet should change by phase and that wearables can forecast next-day blood-sugar instability [14], but these claims have not been tested against hormonally confirmed cycle phases, and their advice is rarely personalized or evidence-labeled.

Goal. Our end goal is to *anticipate next-day symptoms from wearable sensing and respond with personalized, evidence-based meal guidance*. Glucose variability is an *intermediate measure*, not the goal: (1) it is associated with cycle phase and with fatigue and cravings [2], so it may partly explain symptoms; (2) unlike hormones, it can be changed through what and when people eat, so it is the lever meal guidance acts on; (3) few people wear a CGM but many wear a smartwatch, so if a watch can flag likely unstable-glucose days, guidance can reach people without a CGM.

Research questions.

- **RQ1:** Can smartwatch signals and cycle phase forecast next-day symptoms, and which sensors matter most?
- **RQ2:** Does adding glucose improve those forecasts, and can a watch alone flag unstable-glucose days?
- **RQ3:** Can these forecasts drive personalized meal plans through a grounded LLM that is safer and more accurate than an ungrounded one?

Effects may be small in a healthy population, but a negative answer to RQ1 or RQ2 is also informative: it would mean phase- or glucose-based dietary advice is not supported by this data.

2. Literature and Limitations

Glucose across the cycle. Lin et al. [2] combined CGM, Fitbit, and daily hormone tests and found daily median glucose highest in the luteal and lowest in the late-follicular phase, associated with fatigue and cravings. The work is *descriptive*; it does not test whether glucose or symptoms can be *forecast* from wearables.

Wearables for cycle tracking. Wrist skin temperature and other signals predict the fertile window [4, 5], but target *phase or ovulation*, not metabolism or diet.

Modeling on mcPHASES. Recent work classified cycle phase from self-reported symptoms alone (67.6% leave-one-subject-out accuracy), finding that within-person symptom variability carried most of the signal [3]. It did not use metabolic signals; to our knowledge no work tests whether glucose improves symptom forecasting or whether a watch can stand in for a CGM.

Diet, interventions, and LLMs. Trials support ginger for menstrual pain [8], with weaker evidence for other dietary approaches [9]. Just-in-time adaptive interventions (JITAIs) use sensed state to decide when to deliver support [10], but cycle apps rarely link advice to a person's sensed state. LLMs can personalize recipes, but without retrieval grounding [11] they can produce unsupported health claims and wrong nutrition values.

3. Proposed Approach

Data. mcPHASES (PhysioNet, credentialed access): 42 young adults over two $\sim$3-month intervals with Fitbit Sense, Dexcom G6 CGM, Mira urinary hormone kits, and daily self-reports [1]. No new data is collected.

Step 1: Phase alignment (sanity check). Assign phases (menstrual, follicular, fertile, luteal) anchored to the LH surge. With mixed-effects models, replicate the glucose phase pattern of [2] and extend it to variability metrics (coefficient of variation (CV), MAGE, time in range) using CGM consensus definitions [6, 7].

Step 2: Next-day forecasting (RQ1, RQ2; primary contribution).

- *Targets:* (a) next-day fatigue, cramps, bloating, and cravings (primary); (b) a high glucose-variability day, i.e., CV in the top third of that participant's own days.
- *Features (day t predicts $t+1$):* resting and nighttime heart rate (from minute-level data), HRV, skin-temperature deviation, sleep duration and efficiency, activity, and lagged symptoms and glucose, each relative to the participant's own rolling baseline [3].
- *Comparisons:* (1) watch only; (2) watch + cycle phase; (3) watch + phase + glucose. Model (3) vs. (2) answers whether glucose helps; model (2) on target (b) answers whether a watch alone can flag glucose instability. Phase is tested hormone-confirmed (best case) and calendar-estimated (realistic). A *sensor-modality ablation* (heart rate vs. temperature vs. sleep vs. activity) identifies which signals matter, and we report how wear-time gaps affect accuracy.
- *Headline result:* the **deployable configuration**, watch signals plus calendar-estimated phase with no CGM or hormone tests, using only data current smartwatches and period-tracking apps already collect.
- *Models and evaluation:* regularized logistic regression and gradient-boosted trees (no deep models given the sample size) vs. persistence and per-person base-rate baselines; leave-one-subject-out AUROC/AUPRC with bootstrap confidence intervals; SHAP for interpretation. *Exploratory:* mediation analysis of whether glucose variability explains part of the phase–symptom link (associational, not causal).

Step 3: Personalized meal guidance (RQ3). Following the JITAI pattern [10], *sensing decides when and which concern; published evidence decides what to suggest; the LLM personalizes how*. A short in-app preference form collects meals per day, dietary pattern (omnivore, vegetarian, vegan), allergies, cooking time, and an optional height, weight, and goal (maintain, gain, lose) for portion sizing via the Mifflin–St Jeor equation [12], shown as a range, with deficits capped and never below estimated resting needs.

- *Trigger:* three modes keep guidance useful when forecasts are weak. *Proactive*: forecast risk above a confidence threshold triggers next-day guidance. *Phase-based fallback*: otherwise, guidance follows current phase (e.g., iron-rich foods with vitamin C during menses). *Reactive*: a logged symptom triggers immediate guidance.
- *Glucose-aware planning:* on predicted unstable-glucose days, the plan spreads carbohydrates across the user's meals and pairs each with protein or fiber; on other days it is unchanged. This is how the glucose forecast is used.
- *Grounded generation:* retrieval selects candidate recipes from a curated set of about 50 and the matching evidence cards (e.g., ginger when cramps are predicted [8]). The LLM receives the sensed state, forecast, preferences, and retrieved items, and returns a structured meal plan with plain-language explanations that may cite only retrieved evidence. Nutrition is computed in code from USDA FoodData Central [13], never by the LLM, and code checks every plan against the user's restrictions.
- *Evaluation:* 20 synthetic user profiles crossed with real participant days, comparing (a) rule-based ranking, (b) an ungrounded LLM, and (c) our grounded LLM on dietary-restriction violations, unsupported health claims, nutrient error vs. USDA, and whether plans correctly adapt to the forecast.

A dashboard shows the phase timeline, glucose trace, forecasts, and meal plan. Advice is labeled general guidance, not medical advice, with a prompt to see a clinician for severe pain; height and weight stay in the browser. A public demo will use synthetic data, since the data use agreement prohibits redistribution.

4. Potential Challenges

- **Access latency:** PhysioNet credentialing requires CITI training; we start now and prototype on synthetic data.
- **Small *N* and generalizability:** 42 healthy young adults in Canada wearing one device. Results may not transfer to people with PCOS, diabetes, or perimenopause, to hormonal contraception users, or to other watches; we frame findings as a feasibility study.
- **Missing data and irregular cycles:** CGM gaps and incomplete self-reports are handled with per-day coverage rules; cycles without a clear LH surge are analyzed separately.
- **Unvalidated guidance:** mcPHASES has no dietary intervention, so we cannot test whether plans reduce symptoms, and calorie estimates are population-level approximations. We evaluate plan correctness and safety, not health outcomes; testing with real users would require IRB approval and is future work.
- **LLM reliability:** hallucinations and restriction violations are reduced by grounding and code checks, and measured.

5. Planned Timetable (Oct 1 to Dec 11)

Week	Milestone
1 (Oct 1–7)	Submit proposal; start PhysioNet credentialing; finalize literature review
2 (Oct 8–14)	Data access; cleaning; LH-anchored phase alignment; daily CGM metrics
3 (Oct 15–21)	Step 1 replication; curate recipe set and evidence cards (in parallel)
4–5 (Oct 22–Nov 4)	Step 2 models for both targets; preference form and portion calculator
6 (Nov 5–11)	Step 2 ablations (modality, phase, glucose), SHAP, mediation
7 (Nov 12–18)	Step 3 LLM pipeline: retrieval, structured output, restriction checks
8 (Nov 19–25)	LLM evaluation on synthetic profiles; dashboard integration
9 (Nov 26–Dec 2)	Buffer (Thanksgiving week); synthetic demo data; draft report
10 (Dec 3–11)	Final report, presentation, and code release

References

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